

W3 → Random Networks Date

* Random Graph / Network

→ A random graph is a graph of 'N' nodes where each pair of nodes is connected by probability 'p'.

* No. of Links in a Random Network

The Maximum possible number of links = $\frac{N(N-1)}{2}$

$$P(L) = \underbrace{\left[\binom{\frac{N(N-1)}{2}}{L} \right]}_{\text{No. of ways we can choose } L \text{ links from all possible links.}} p^L (1-p)^{\frac{N(N-1)}{2} - L}$$

No. of ways we can choose L links from all possible links.

Probability of getting exactly n successes

$$P(n) = \underbrace{\binom{N}{n}}_{\text{choosing } n \text{ successes}} p^n \underbrace{(1-p)^{N-n}}_{\text{probability of failure.}} \rightarrow \text{Probability of failure.}$$

Arg value (mean)

$$\langle n \rangle = Np$$

Ex:- 10 trial, $p = 0.3$

$$\text{Expected Success} = 10 \times 0.3 = 3$$

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* Probability of having exactly 'L' links

$$P(L) = \left[\binom{\frac{N(N-1)}{2}}{L} p^L (1-p)^{\frac{N(N-1)}{2} - L} \right]$$

→ Total Possible links = $\frac{N(N-1)}{2}$

→ choosing L links from the total possible links

$$\langle L \rangle = p \frac{N(N-1)}{2} \rightarrow \text{Avg. Number of links}$$

$$\langle K \rangle = \frac{2L}{N} = p(N-1) \rightarrow \text{Avg. Degree of a Node}$$

Random Networks have Binomial Distribution approximating to Poisson for $K \ll N$

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* Degree Distribution of a Random Graph

→ Degree distribution is the probability that a node has exactly K connections.

$$P(K) = \binom{N-1}{K} p^K (1-p)^{(N-1)-K}$$

Binomial Dist.

selecting K Nodes from $(N-1)$ nodes

probability of having K edges

probability of missing $(N-1)-K$ edges.

Average Degree

$$\langle K \rangle = p(N-1)$$

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* Poisson Degree Distribution

$$P(R) = \binom{N-1}{k} p^k (1-p)^{(N-1)-k} \quad \left. \vphantom{\binom{N-1}{k}} \right\} \begin{array}{l} \text{Binomial} \\ \text{Distribution} \end{array}$$

Average Degree

$$\langle k \rangle = p(N-1)$$

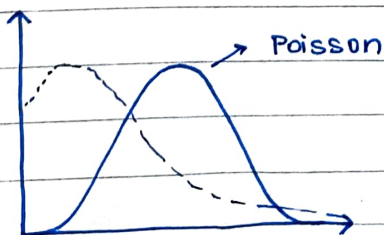
For large 'N' & small 'k', we arrive to Poisson Distribution

$$P(k) = e^{-\langle k \rangle} \frac{\langle k \rangle^k}{k!}$$

* Real Networks are Not Poisson

~~* Real Networks are supercritical~~

- Many Nodes have few connections
- Few Nodes have very large no' of connections (hubs)



Small world Phenomenon / Six degrees of separation

→ Two Individuals anywhere in the world can be connected by a chain of six or less ~~steps~~ ^{acquaintances} Date.....

* Real Networks are supercritical

→ one giant connected component exists

→ Most Nodes belong to one big Network

→ You can reach any Node from any other Node.

* Distances in Random Graphs

$\langle K \rangle$ nodes at distance one ($d=1$)

$\langle K \rangle^2$ " " " two ($d=2$)

1

1

$\langle K \rangle^d$ " " " d ($d=d$)

$$\text{Total Node (N)} = 1 + \langle K \rangle + \langle K^2 \rangle + \dots + \langle K \rangle^{d_{\max}}$$

$$= \frac{\langle K \rangle^{d_{\max}+1} - 1}{\langle K \rangle - 1} \cup \langle K \rangle^{d_{\max}}$$

∴

$$d_{\max} = \frac{\log N}{\log \langle K \rangle}$$

$$\langle d \rangle \approx \frac{\log N}{\log \langle K \rangle}$$

* More connections (larger K) → shorter distances

Denser Network → smaller dist b/w Nodes

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* clustering coefficient

$$C_i = \frac{2 L_i}{K_i (K_i - 1)}$$

$K_i \rightarrow$ degree of Node i
 $L_i \rightarrow$ No. of links b/w neighbours of node i .

$$C_i = \frac{\text{actual links}}{\text{max-possible links}} = \frac{L_i}{\frac{K_i (K_i - 1)}{2}}$$

clustering in Random graphs

$\rightarrow$ Each edge exists independently with a probability p

$\therefore$ Expected No. of Links among Neighbours $\langle L_i \rangle$

$$\langle L_i \rangle = p \frac{K_i (K_i - 1)}{2}$$

$$\therefore C_i = \frac{2 \langle L_i \rangle}{K_i (K_i - 1)} = p = \frac{\langle K \rangle}{N} \quad \therefore C_i = \frac{\langle K \rangle}{N}$$

1) clustering coeff. of random graphs is small

2) for fixed $\langle K \rangle$ clustering $(C_i) \downarrow$, if $N \uparrow$

3) clustering is independent of the Node degree K .

$\{$ whether a node has 10 neighbours / 100 neighbours, the probability of links b/w them stays the same i.e. p $\}$.

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$$c_i = \frac{2 \langle L_i \rangle}{k_i(k_i - 1)} = p = \frac{\langle K \rangle}{N}$$

→ c decreases with the system size N .

→ c is independent of Node's degree K .

* Watts - Strogatz Model

→ Before this paper / Model, two main types of network models existed :-

1) Regular Networks

- Nodes connected only to nearby neighbours { Ex - Ring Lattice }
- High clustering (neighbours are connected)
- High Path length (far nodes are many steps away)

2) Random Networks

- connections placed randomly
- Low clustering
- Low Path length

Most Real Networks have :-

- High clustering
 - low Path length
- } Small World Networks

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Watts & Strogatz Model :-

→ start with a regular Network (High clustering)

→ Randomly rewire some edges with probability p .
(As $nK/2$ edges, rewiring stops after $K/2$ laps)

Regular Network $\rightarrow$ Small-world $\rightarrow$ Random Network
($p=0$) (small p) ($p=1$)

* observations

- 1) Path length dropped very fast
 - 2) clustering remained high.
- } Small world Network.

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* Real Networks are not Random

comparing Real Networks with Random graphs on the basis of:-

1) Path lengths (✓)

Random graph prediction:- $\langle d \rangle \approx \frac{\log N}{\log \langle k \rangle}$

→ Real Networks also have short distances like the random graphs.

2) clustering co-efficient (X)

Random graph prediction:- $c_i = \frac{2 \langle L_i \rangle}{k_i(k_i - 1)} = p = \frac{\langle k \rangle}{N}$

The above tells, clustering becomes small as Network grows.

Real Networks have :-

→ clustering much higher than predicted.

→ Ex:- friends of friends are usually friends.

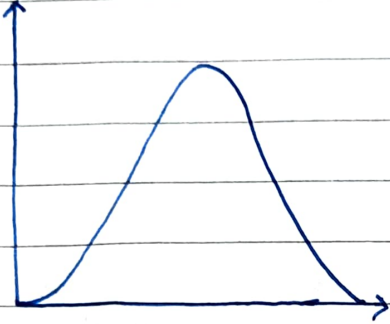
∴ $c_{rand} \ll c_{real}$

→ c_{rand} underestimates the clustering coefficient of real Networks

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3) The Degree Distribution (X)

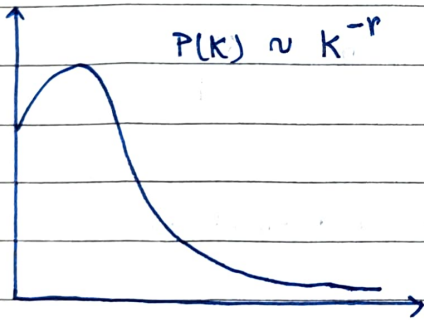
Prediction $\rightarrow P(k) = e^{-\langle k \rangle} \frac{\langle k \rangle^k}{k!}$ } Poisson Distribution



$\rightarrow$ Most Nodes have degree near the Avg.

$\rightarrow$ Hubs are rare

observed



$$P(k) \sim k^{-\gamma}$$

$\rightarrow$ Power law distribution

$\rightarrow$ Many Small degree Nodes

$\rightarrow$ Few Extremely large Hubs.

∴ Are Real Networks Random? No!!